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Research Paper

Computational study of single-phase immersion cooling for high-energy density server rack for data centers

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ABSTRACT

As the cooling difficulty and power consumption of data centers increase in line with the growing demand for high-performance data centers, the importance of thermally effective and energy-efficient cooling of data centers has become apparent. However, the research on managing high heat generating electronic components in a server rack remains insufficient, failing to address the recent developments in the high-energy density server racks. This study narrows that gap through the computational analysis of the cooling systems for the server rack. Considering the varying impacts of different parameters of the cooling methods, this study comprehensively analyzed immersion cooling on a high-energy density server rack with high-performance components by comparing multiple cooling methods. Under equal power consumption, immersion cooling can achieve the lowest component temperature with the highest stability at various central processing unit (CPU) and memory thermal design powers, power consumptions, inlet temperatures, and server heights. For 800 W CPUs and 20 W memories, immersion cooling is the only method capable of maintaining component temperatures below the target temperature in an extreme condition, maintaining a CPU and memory temperature of 62.0 °C and 44.2 °C. Additionally, the temperature variation of the CPU was the lowest at 9.1 K, while those for air and hybrid cooling were 41.6 K and 10.7 K, respectively. The impact of various dielectric fluids on the component temperature for immersion cooling was investigated with a focus on the thermal properties of the fluid along with the thermal and hydraulic characteristics of the heatsink simplified through a porous media model.

1. Introduction

The rapid development of information technology (IT), encompassing big data, artificial intelligence (AI), and the Internet of Things, has significantly enhanced the value of data centers, which are capable of processing and storing vast quantities of information [1]. The number of operational high-performance hyperscale data centers has increased rapidly, attaining 700 by 2021 [2]. Furthermore, the amount of energy consumed by the data centers is projected to grow from 286 TWh in 2016 to 752 TWh in 2030, largely due to the high power consumption in the cooling system [3]. To address the high energy consumption of data centers, various cooling methods, including air cooling, indirect liquid cooling, and immersion cooling, are employed with thermal effectiveness also in mind. Air cooling, the most common method, utilizes fans to blow air, a primary coolant, over the electronic components [4]. Indirect liquid cooling, on the other hand, uses liquids with higher densities and thermal conductivities than air. The coolant flows through a cold plate attached to heat-generating devices, such as central processing units (CPUs), and cools them indirectly [5]. This method is typically incorporated with the air cooling for the thermal management of other components, hence the term "hybrid cooling." In contrast, immersion cooling, also known as direct cooling, employs a dielectric fluid that is in direct contact with the servers. This fluid does not conduct electricity under normal conditions, with a breakdown voltage as high as 46 kV for FC [6].

With the advantages of energy savings, cost-effectiveness, and compact design, immersion cooling has emerged as the primary field of research in server cooling [7]. Hu and David (2024) conducted a simulational analysis of three dielectric fluids for passive immersion cooling [8]. Without forced convection considered in the model, the Rayleigh number was the driving non-dimensional term in the thermal performance. In addition, addressing the importance of the thermophysical properties of the coolant, Birbarah et al. (2020) coated the server with Parylene C for electrical insulation and immersed the server in water with a higher thermal conductivity than many dielectric fluids [9]. Focusing on temperature uniformity across the server rack and its distinction from a single server, Liu and Chang (2024) conducted a

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Nomenclature

Symbols

b	base thickness [m]
Br	Brinkman number [-]
C	inertial resistance factor [m^{-1}]
c_p	specific heat capacity [$m^2 \cdot s^{-2} \cdot K^{-1}$]
c_1	coefficient for pressure drop equation of heatsink [$kg \cdot m^{-2} \cdot s^{-1}$]
c_2	coefficient for pressure drop equation of heatsink [$kg \cdot m^{-3}$]
D	diameter [m]
dT	temperature difference [K]
F	force [$kg \cdot m \cdot s^{-2}$]
f	friction factor [–]
g	gravitational acceleration [$m \cdot s^{-2}$]
h	fin height [m]
K	loss coefficient [–]
k	thermal conductivity [$kg \cdot m \cdot s^{-3} \cdot K^{-1}$]
L	length of heatsink [m]
L^*	dimensionless length of heatsink [–]
N	number of fins [–]
n	number of CPUs [–]
P	pressure [$kg \cdot m^{-1} \cdot s^{-3}$]
Q	heat transfer rate [$kg \cdot m^2 \cdot s^{-3}$]
Re	Reynolds number [–]
S	momentum source [$kg \cdot m \cdot s^{-1}$]
T	temperature [K]
t	fin thickness [m]
u	velocity [$m \cdot s^{-1}$]
$\dot{V}$	volumetric flow rate [$m^3 \cdot s^{-1}$]
W	width of heatsink [m]
w	width [m]
x	thickness of medium [m]

Greek symbols

α	permeability [m^2]
μ	dynamic viscosity [$kg \cdot m^{-1} \cdot s^{-1}$]
ν	kinematic viscosity [$m^2 \cdot s^{-1}$]
ϕ	porosity [–]
ρ	density [$kg \cdot m^{-3}$]
σ	fraction of frontal free flow area [–]

Subscripts

amb	ambient
app	apparent
avg	average
c	contraction
ch	channel
d	drag
e	expansion
eff	effective
f	fluid
h	hydraulic
i	i-direction
in	inlet
max	maximum
min	minimum
s	solid
std	standard deviation

Abbreviations

AI	artificial intelligence
CFD	computational fluid dynamics
CPU	central processing unit
IT	information technology
MO	mineral oil
NE	natural ester
PC	power consumption [$kg \cdot m^2 \cdot s^{-3}$]
PCB	printed circuit board
PG	propylene glycol
TDP	thermal design power [$kg \cdot m^2 \cdot s^{-3}$]
TIM	thermal interface material
TR	thermal resistance [$kg^{-1} \cdot m^{-2} \cdot s^3 \cdot K$]

multi-objective optimization [10]. Through Taguchi-based relational analysis, they identified the contribution of various structural parameters to the performance index, with the baffle angle emerging as the most important. Liu et al (2024) also focused on the configuration of the server rack tank for the uniform flow across the server rack [11]. With various parameters including the porous panel at the bottom of the tank, the tank design was optimized for the lowest temperature, pressure drop, and coolant weight. Along with the importance of the immersion tank design, the importance of the heatsink design was noted by Herring et al. (2024), as the novel additive manufacturing technology is used to create a body-centered cubic lattice design for the heatsink [12]. Both the thermal and pressure drop characteristics of the heatsink were studied for the development of an optimized design. In addition, various novel systems were integrated with the immersion cooling for further improvements. For two-phase (2P) immersion cooling, incorporating phase change of the dielectric fluids for higher heat transfer coefficients, a vapor chamber heat spreaders were attached to the electronic components for better and stable dissipation of heat [13]. Additionally, a jet liquid cooling system with a high velocity stream of liquid directed to the heat generating chips is integrated with immersion cooling for better cooling performance with additional power consumption [14]. Similarly, the targeted flow of the dielectric fluid directly through the heatsink attached to the CPU is studied for a more efficient flow in the immersion cooling system [15].

In addition to these advancements, previous research has compared various cooling methods under different conditions, as depicted in Table 1. While some comparisons include immersion cooling, others focus solely on air or hybrid cooling methods, highlighting the diversity of approaches studied in the fields. Ramakrishnan et al. (2021) tested the overclocking performance of air cooling, hybrid cooling, and 2P immersion cooling for the maximum computing performance of a CPU with thermal design power (TDP) of 235 W [16]. Two-phase immersion cooling demonstrated the best performance under harsh conditions, with a core frequency 51 % higher than that of air cooling. Wang et al. (2024) compared the temperatures of various components including the CPU and memory for air, hybrid, and immersion cooling under identical mass flow rates [17]. The CPU temperature with 200 W TDP for immersion cooling was 34.8°C and 20.8 °C lower than that for air and hybrid cooling, respectively, and immersion cooling exhibited the most uniform coolant temperature and low power consumption. Zimmermann et al. (2012) focused on the energy efficiency and exergy analysis to compare the performances of air and hybrid cooling [18]. The hybrid cooling system was observed to be capable of successfully managing the temperature of the components using a hot coolant of 60°C, which enabled the efficient reuse of the heat and yielded a higher exergy output. Parida et al. (2012) conducted experiments with a cold plate or cold rail attached to memories and CPUs for comparison with air cooling [19]. Their findings indicated that cold plates could save energy while demonstrating a higher thermal performance for both memories and CPUs. Almoli et al. (2012) adopted a broad approach to compare air and

Table 1
Literature review on comparison of data center cooling methods.

Ref.	Author (Year)	Maximum CPU TDP (W)	Server configuration	Cooling method			Dielectric fluid
				Air	Hybrid	Immersion	
[16]	Ramakrishnan et al. (2021)	235	Single	✓	✓	✓(2P)	HFE-7000
[17]	Wang et al. (2024)	200	Single	✓	✓	✓	Mineral oil, Synthetic oil Silicone oil
[18]	Zimmermann et al. (2012)	90	Rack	✓	✓		
[19]	Parida et al. (2012)	130	Single	✓	✓		
[20]	Almoli et al. (2012)	–	Rack	✓	✓		
[21]	Sahini et al. (2017)	130	Single	✓	✓		
[22]	Gandhi et al. (2019)	95	Single	✓		✓	EC-110, Mineral oil
[23]	Murthy et al. (2022)	200	Single	✓		✓	EC-110
[24]	Chi et al. (2014)	115	Rack		✓	✓	Fluoro-organic
–	Current study	800	Rack	✓	✓	✓	FC-40, Novec 7500, TMC-7200 SF 10, S5 X, EC-110, PAO 4, Mineral oil, Natural ester

hybrid cooling by utilizing a computational fluid dynamics (CFD) model of a data center with three aisles and six racks [20]. Through liquid cooling, the heat load of an air conditioning system reduces significantly owing to the high specific heat capacity of the liquid. Sahini et al. (2017) measured the CPU temperature under a wide range of coolant inlet temperatures to compare air and hybrid cooling [21]. Even with fans consuming more power, the CPU temperature with hybrid cooling at an inlet temperature of 45 °C was 18 °C lower than that with air cooling. Gandhi et al. (2019) employed the immersion cooling method to compare the temperatures of components with those for air cooling [22]. At identical coolant inlet temperatures, immersion cooling was demonstrated to be more effective for cooling the CPU and various chipsets. Murthy et al. (2022) demonstrated the effectiveness of immersion cooling in terms of thermal management of components and power consumption [23]. Chi et al. (2014) experimentally compared the energy efficiencies of hybrid and immersion cooling data centers with a facility capacity of 250 kW [24]. The results indicated that the hybrid cooling system required 48 % of the IT load for cooling which was significantly higher than the 14 % load required for immersion cooling.

Throughout the literature review, various cooling methods such as air, hybrid, and immersion cooling were evaluated for their effectiveness in managing the component temperature and power consumption. However, the range of the CPU TDP was only limited from 90 W to 235 W. Considering the recent advancements in CPU technology and the potential future demand for more powerful CPUs, the scope of the previous study was limited [25]. Furthermore, these comparisons were primarily conducted on the cooling performance of a single server with a limited number of components failing to consider the interaction between the heat generating components in a rack. Limited research utilizing server racks focused on the energy efficiency of data center buildings rather than the thermal management of individual components. As depicted in the literature review of the immersion cooling, uniformity of the component temperature and the fluid flow is a key performance index in the application of the server rack. However, the comparison of the cooling methods in terms of the uniformity across the server rack was limited. Finally, most studies were conducted under a limited number of operating conditions, failing to capture the weight of these variables on different cooling methods. As a result, there is a need for a more comprehensive study of the cooling methods that accounts for the wider scope.

To address the limitations of previous studies, the immersion cooling method along with other methods were analyzed exhaustively for a high-energy density server rack with a high CPU TDP. The potential performance enhancements were considered through a comparison with other cooling methods. In addition, four single servers were stacked on top of each other to provide a realistic representation of an actual high-energy density server rack during operation. The study of data center cooling as a rack allows for the evaluation of cooling methods in terms of the temperature uniformity of the CPUs across the server rack, which is indicative of stability. In addition, through various operating conditions, the distinct significance of each variable in three different cooling methods could be identified. For immersion cooling, this study builds upon and extends previous work by addressing the importance of heatsink and thermophysical properties of various dielectric in terms of the thermal and hydraulic characteristics.

To achieve these objectives, a computational approach was adopted, with the simulation model simplified by utilizing a porous media model for the heatsink. The model was validated with experimental data from the literature. With the developed model, immersion cooling was analyzed thoroughly through a comparison with other cooling methods under various conditions, including the CPU TDP, memory TDP, power consumption, coolant inlet temperature, and server height, for the high-performance server configuration. To gain further insights into the performance of immersion cooling under extreme conditions, the cooling performances of various dielectric fluids were evaluated with a particular focus on their thermal properties.

2. Research approach

2.1. Design description

In this study, the Cisco UCS C220 M3 rack server was selected as the model for comparing air, hybrid, and immersion cooling as depicted in Fig. 1. The height of the server is 1 U or one rack unit, which is defined as 44.45 mm. The server comprises two CPUs, 16 memories, a power supply unit, a printed circuit board (PCB), a platform controller hub, and other components. Sahini et al. [21] employed such a server in an experiment on the thermal management of CPU and memory by using air and hybrid cooling methods. The experimental data from this paper was subsequently used to validate a three-dimensional (3D) simulation model developed using Ansys Fluent 23.1 software. With the server model as a baseline, Fig. 1 illustrates four rack servers placed on top of each other, which depicts servers placed in a rack. This approach allows for the investigation of the thermal impact of a server on another and thereby enables a more in-depth analysis of data center cooling.

The air cooling depicted in Fig. 2**(a)** utilizes heatsinks and fans to enhance the heat dissipation of the CPU and create airflow through the heatsinks and other components. Heatsinks with the configurations depicted in Table 2 are made of copper with high thermal conductivity, as detailed in Table 3. Heatsinks are attached to the CPUs that generate the most heat, thereby increasing the surface area through the fins to facilitate heat removal. A thermal interface material (TIM) is placed between a heatsink and CPU to reduce thermal resistance by removing air gaps that interfere with the heat conduction between these. For the airflow through the server, five Delta DC fans (model GFB0412EHS-DF00) were used, and these were modeled utilizing a performance

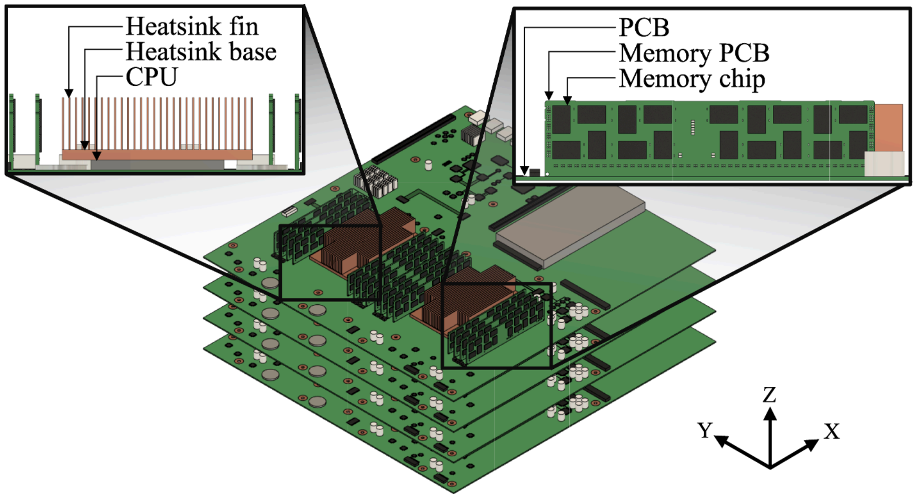

Fig. 1. Isometric view of 4 × 1 U rack server configuration.

curve relating the air flow rate and static pressure. The thermal properties of the air and the other fluids are listed in Table 4. Unlike the properties of solids in Table 3, the properties of gases and liquids are expressed as functions of temperature owing to the wider range of properties and for a more accurate depiction of the actual phenomenon.

Another approach to server cooling, illustrated in Fig. 2**(b)**, is hybrid cooling, which employs a cold plate for the thermal management of CPUs and conventional air cooling for other components such as memories. Unlike air cooling, the CPU heatsink is replaced with a copper cooling block containing a serpentine fluid channel, as depicted in Fig. 3. The cold plates consist of a copper block enclosed within a plastic casing, connected by rubber tubing with an internal diameter of 5 mm. The coolant is propylene glycol (PG) solution (35 % propylene glycol and 65 % di-water by volume) which flows through the copper block to cool two CPUs in a single path. Despite the change in the CPU cooling, the thermal management of the other components is still achieved through the fans like air cooling. The thermal properties of the materials used in hybrid cooling are listed in Table 3 and Table 4.

Finally, immersion cooling is modeled in Fig. 2**(c)** for a comparison with air and hybrid cooling. Immersion cooling is a novel cooling method in which an entire server is submerged in a dielectric fluid. For this research, FC-40 with a high density and low viscosity was selected as the working fluid among various types of dielectric fluids listed in Table 5. Like air cooling, heatsinks are attached to the CPUs to lower their temperature through a larger surface area, as depicted in Fig. 4. Considering the high viscosity and higher heat transfer coefficient of the dielectric fluid compared with air, certain modifications were made to the configurations of the heatsinks, as shown in Table 2. The width of the channel was widened to facilitate a better flow of the dielectric fluid with high viscosity. In addition, the width of the fin was also increased to accommodate a higher heat transfer coefficient. As the working fluid was changed from air to dielectric fluid, the fans were replaced with pumps for the flow through the server. Furthermore, the inlet–outlet layout was changed to a T-configuration with three inlets and two outlets in a T-shape, which demonstrated superior performance compared with the other layouts [35].

2.2. Governing equation

To simplify the simulation model and the depict actual situations realistically, the following assumptions were made for the 3D CFD model:

- The flow was considered steady and incompressible.
- The viscous dissipation term and radiation heat transfer were considered negligible due to the low Brinkman number and specific conditions of the study.
- The temperatures of the solid and the fluid in the porous media were considered identical (local thermal equilibrium was assumed).
- The CPU and memory chips were assumed to generate uniform heat across the volume.
- The gravitational effect was considered in the negative x-direction for immersion cooling and the negative z-direction for air and hybrid cooling, as illustrated in Fig. 1.
- The porous media were considered anisotropic to accurately model the flow along the channels while accounting for the restricted flow across them.

Brinkman number, calculated using **Eqs. (1),** was employed to assess the significance of the viscous dissipation term in this study. Since the Brinkman number was far below the threshold value of 1, viscous dissipation was considered negligible.

$$\mathrm{Br} = \frac{\mu u^2}{k(T_{wall} - T_{in})} \tag{1}$$

The governing equations for continuity, momentum, and energy are given as **Eqs. (2)–(5)** for the entire simulation domain except the porous media zone.

$$\nabla \bullet \left(\rho_f \vec{u}_f\right) = 0 \tag{2}$$

$$\nabla \bullet \left(\rho_f \vec{u}_f\right)\vec{u}_f = \nabla \bullet \left(\mu_f \nabla \vec{u}_f\right) + \rho_f \vec{g} - \nabla P \tag{3}$$

$$\frac{\partial}{\partial t}(\rho_f c_{p,f} T_f) + \nabla \bullet \left(\rho_f c_{p,f} T_f \vec{u}_f\right) = \nabla \bullet (k_f \nabla T_f) \tag{4}$$

$$\nabla \bullet (k_s \nabla T_s) = 0 \tag{5}$$

For the heatsinks employed in air and immersion cooling, plate-fin heatsinks were modeled as simple blocks using an equivalent porous media model [43]. Modeling heatsinks with exceptionally thin fins can result in an excessive mesh quantity and a longer computational time. To reduce the computational cost significantly, a porous media model was employed, wherein the heatsinks were assumed to be porous media consisting of pores through which fluids can flow. In this method, additional inputs (porosity, permeability, and inertial resistance factor of the heatsink) are required for the computation. To calculate the

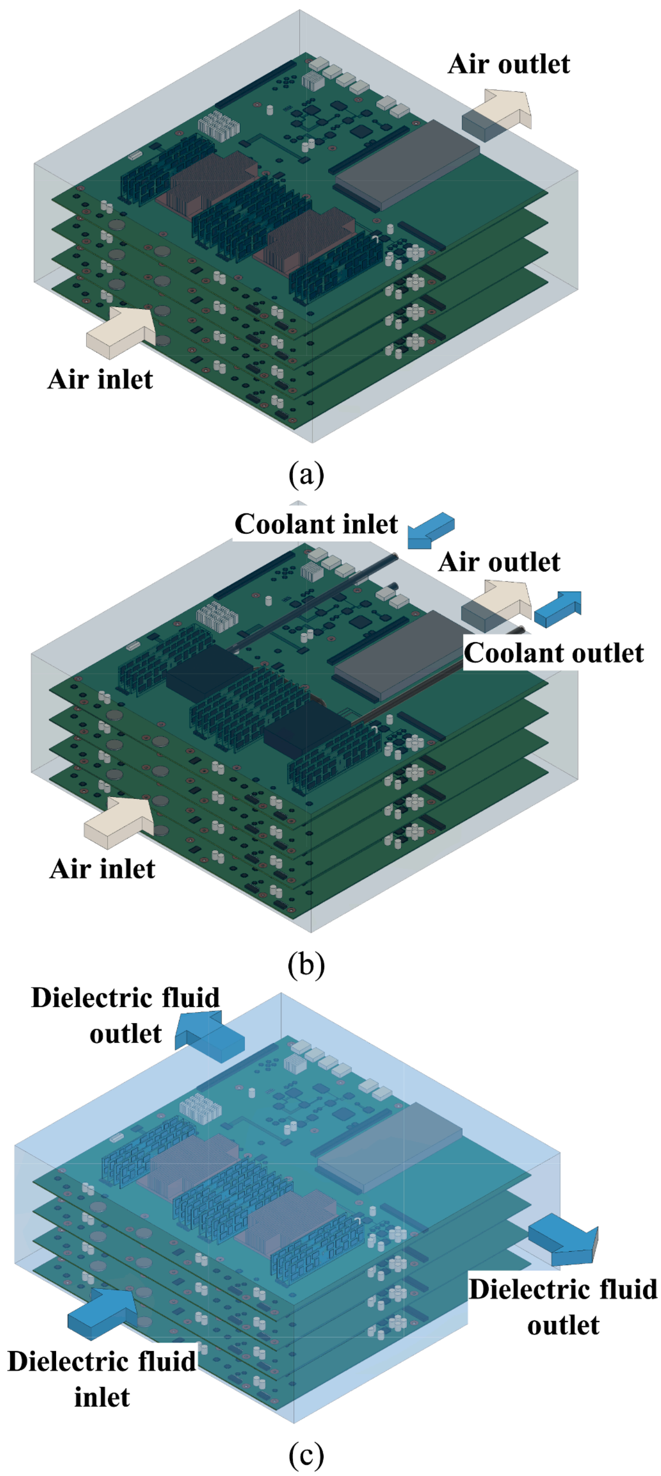

Fig. 2. Schematic diagram of (a) air, (b) hybrid, and (c) immersion cooling server.

permeability and inertial resistance factor required for the additional momentum sink term in the momentum governing equation for the porous media, the pressure drop across the inlet and outlet of the heatsink was computed first using **Eqs.** (6)–(16). These equations account for the friction factor for the developing and fully developed laminar flow across a rectangular channel, the expansion loss coefficient, and the contraction loss coefficient [44]. From these equations, the pressure drop for various flow velocities can be expressed as a second-order polynomial with a y-intercept of zero.

$$\frac{F_{\mathrm{d}}}{\left(\frac{1}{2}\right)\rho u_{\mathrm{ch}}^{2}} = f_{\mathrm{app}}N(2hL + w_{\mathrm{ch}}L) + K_{\mathrm{c}}(hW) + K_{\mathrm{e}}(hW) \tag{6}$$

$$\Delta P_{\mathrm{heatsink}} = \frac{F_{\mathrm{d}}}{hW} \tag{7}$$

$$u_{\mathrm{ch}} = u_{\mathrm{f}}\left(1 + \frac{t}{w_{\mathrm{ch}}}\right) \tag{8}$$

$$f_{\mathrm{app}}Re_{D_{\mathrm{h}}} = \left[\left(\frac{3.44}{\sqrt{L^{*}}}\right)^{2} + \left(fRe_{D_{\mathrm{h}}}\right)^{2}\right]^{\frac{1}{2}} \tag{9}$$

$$Re_{D_{\mathrm{h}}} = \frac{D_{\mathrm{h}}u_{\mathrm{ch}}}{\nu} \tag{10}$$

$$D_{\mathrm{h}} = 2w_{\mathrm{ch}} \tag{11}$$

$$L^{*} = \frac{L}{D_{\mathrm{h}}Re_{D_{\mathrm{h}}}} \tag{12}$$

$$fRe_{D_{\mathrm{h}}} = 24 - 32.527\left(\frac{w_{\mathrm{ch}}}{H}\right) + 46.721\left(\frac{w_{\mathrm{ch}}}{H}\right)^{2} - 40.829\left(\frac{w_{\mathrm{ch}}}{H}\right)^{3} + 22.954\left(\frac{w_{\mathrm{ch}}}{H}\right)^{4} - 6.089\left(\frac{w_{\mathrm{ch}}}{H}\right)^{5} \tag{13}$$

$$K_{\mathrm{c}} = 0.42\left(1 - \sigma^{2}\right) \tag{14}$$

$$K_{\mathrm{e}} = \left(1 - \sigma^{2}\right)^{2} \tag{15}$$

$$\sigma = 1 - \frac{Nt}{W} \tag{16}$$

The governing equations of continuity for porous media (**Eq.** (17)) were employed, which is identical to **Eq.** (2) for the steady state. For momentum, the calculated pressure drop was utilized for an additional momentum sink term in **Eq.** (18). From the pressure drop, the Darcy-Forchheimer equation at **Eq.** (19) with **Eq.** (20)–(22) was utilized to

Table 3
Properties of solids.

Material		Density ($kg \cdot m^{-3}$)	Thermal conductivity ($W \cdot m^{-1}K^{-1}$)	Specific heat ($J \cdot kg^{-1}K^{-1}$)
Copper	[26]	8,978	387.6	381
PCB	[27]	3,600	42.0 (In-plane)0.5 (Cross-plane)	720
TIM	[28]	2,600	3.0	1,200
CPU	[29]	2,300	98.4	721
Casing	[30]	1,380	0.16	900
Rubber	[31]	850	0.25	2,217

Table 2
Configuration of heatsink.

Type	w_{ch} (mm)	t (mm)	h (mm)	b (mm)	L (mm)	W (mm)	N	ϕ
Air cooling	1.60	0.22	20.6	4.00	123	75.0	40	0.88
Immersion cooling	1.97	0.60	20.6	4.00	123	75.0	30	0.77

Table 4
Properties of gas/liquids.

Material		Density (kg·m^{-3})	Thermal conductivity (W·m^{-1}K^{-1})	Specific heat (J·kg^{-1}K^{-1})	Viscosity (10^{-3} kg·m^{-1}s^{-1})
Air	[32]	$0.553 + 8.19 \cdot 10^{-3} \cdot T - 2.04 \cdot 10^{-5} \cdot T^2$	$0.256 – 1.60 \cdot 10^{-3} \cdot T^2 + 2.76 \cdot 10^{-6} \cdot T^3$	$-41800 + 408 \cdot T - 1.30 \cdot T^2 + 1.38 \cdot 10^{-3} \cdot T^3$	$0.290 – 2.79 \cdot 10^{-3} \cdot T + 9.43 \cdot 10^{-6} \cdot T^2 - 1.05 \cdot 10^{-8} \cdot T^3$
PG solution	[33]	$122 – 0.65 \cdot T$	$0.346 + 3.20 \cdot 10^{-4} \cdot T$	$3090 + 2.47 \cdot T$	$132 – 0.750 \cdot T + 1.08 \cdot 10^{-3} \cdot T^2$
FC-40	[34]	$2499 – 2.16 \cdot T$	$0.086 – 6.90 \cdot 10^{-5} \cdot T$	$590 + 1.55 \cdot T$	$42.9 – 0.162 \cdot T + 1.08 \cdot 10^{-4} \cdot T^2$

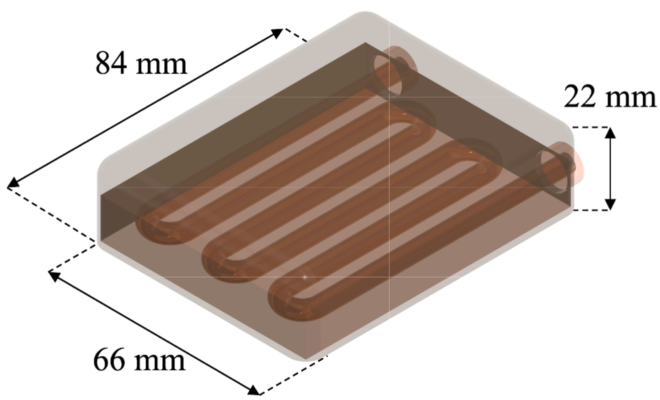

Fig. 3. Detailed schematic of the cold plate in hybrid cooling.

Table 5
Properties of various dielectric fluids.

Material		Density (kg·m^{-3})	Thermal conductivity (W·m^{-1}K^{-1})	Specific heat (J·kg^{-1}K^{-1})	Viscosity (10^{-3} kg·m^{-1}s^{-1})
FC-40	[6]	1870.1	0.066	1042.0	4.9
TMC-7200	[36]	1430.0	0.075	1220.0	0.6
SF10	[37]	1591.8	0.077	1240.0	1.3
Novec 7500	[38]	1628.3	0.066	1118.0	1.3
S5 X	[39]	806.0	0.142	2274.0	7.9
EC-110	[40]	821.2	0.172	2128.9	14.0
PAO 4	[41]	819.0	0.153	2042.1	17.6
Mineral oil	[42]	871.3	0.134	1862.5	19.6
Natural ester	[42]	922.1	0.183	1993.7	67.0

derive permeability and inertial resistance factor required for the input in the CFD simulation.

$$\nabla \bullet \left(\rho_f \vec{u}_f\right) = 0 \tag{17}$$

$$\nabla \bullet \left(\rho_f \vec{u}_f\right) \vec{u}_f = \nabla \bullet \left(\mu \nabla \vec{u}_f\right) + \rho_f \vec{g} - \nabla P - S_i \tag{18}$$

$$\frac{\Delta P_{heatsink}}{\Delta x_i} = S_i = \frac{\mu}{\alpha} u_i + \frac{1}{2}\rho C u_i^2 \tag{19}$$

$$\Delta P_{heatsink} = c_1 u_i + c_2 u_i^2 \tag{20}$$

$$\frac{1}{\alpha} = \frac{c_1}{\mu \Delta x_i} \tag{21}$$

$$C = \frac{2c_2}{\rho \Delta x_i} \tag{22}$$

As the local thermal equilibrium or an equal fluid and solid temperature is assumed with the limited temperature difference between the fluid and solid regions in the porous media zone, **Eq.** (23) was used for the governing equation of energy. The effective thermal conductivity of the porous media zone was calculated using the thermal conductivities of the fluid and solid with porosity as shown in **Eq.** (24).

$$\nabla \bullet \left(\rho_f c_{p,f} \vec{u}_f T\right) = \nabla \bullet \left(k_{eff} \nabla T\right) \tag{23}$$

$$k_{eff} = \phi k_f + (1 - \phi) k_s \tag{24}$$

Utilizing Ansys Fluent 23.1 software, these governing equations were solved numerically with the pressure-based solver at a steady state with the assumption of a steady flow of the fluids and heat generation of electronic components. For the fluids in the simulation model, velocity inlet and pressure outlet boundary conditions were utilized with identical inlet fluid temperatures for three cooling methods. For the

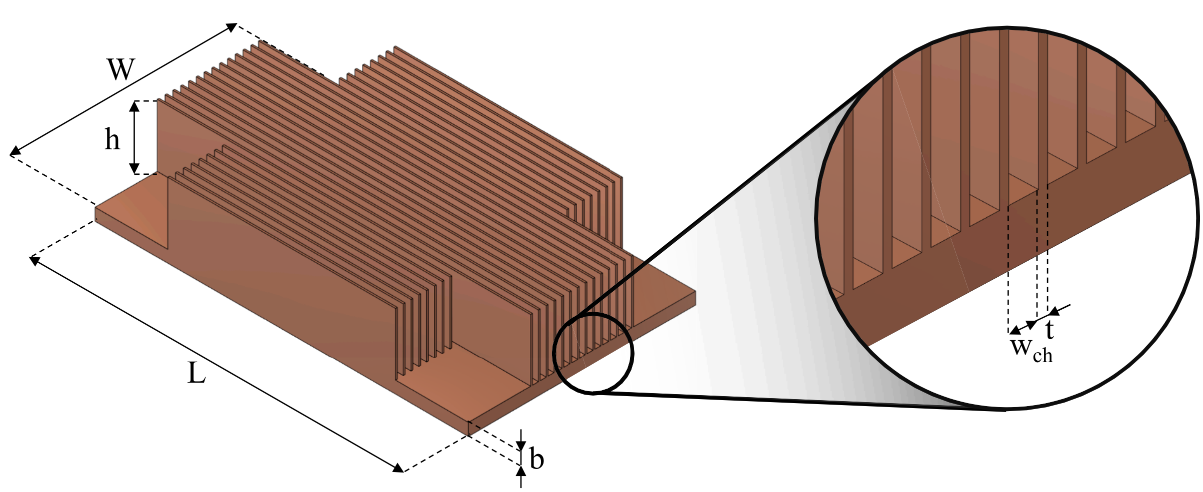

Fig. 4. Plate-fin heatsink for immersion cooling.

boundary condition of the outer surface of the server which is in direct contact with the ambient air during the operation, constant natural convection heat transfer coefficient was assumed. A coupled scheme was employed for pressure–velocity coupling. For the specified simulation model, convergence criteria for residual were set as 10^{-3} for continuity and momentum equations along with 10^{-6} for energy equation during the iteration.

2.3. Grid independence analysis

A grid independence analysis was conducted to test the sensitivity of the simulation results to the number of grids. As the mesh becomes finer, the simulation results become less dependent on the number of grids and converge. However, this results in inefficiency in terms of computational time. Meanwhile, as the mesh becomes coarser, the output of the simulation may vary significantly depending on the mesh size, and the results may be inaccurate. Therefore, it is fundamental to select a suitable mesh size to minimize the computational cost and obtain accurate results. As illustrated in Fig. 5, a grid independence analysis was conducted for a server utilizing air cooling with the following performance indexes: the average CPU temperature and air mass flow rate. A wide range of meshes, spanning from 330,868 to 26,464,288, was used for the simulation, and the mesh size was selected such that the deviations in the CPU temperature and air mass flow rate were within 0.1 K and 0.5 %, respectively. At a grid number of 8,294,418, the criterion was satisfied with a temperature deviation of 0.007 K and mass flow rate deviation of 0.037 %.

2.4. Model validation

A CFD tool was employed to develop simulation models, which were validated subsequently against the experimental results for air and hybrid cooling under various boundary conditions employed by Sahini et al. [21]. A Cisco UCS C220 M3 rack server with two CPUs of 135 W TDP was cooled through air and hybrid cooling methods, and the average temperatures of these CPUs were measured and used for validation purposes. Seven experiments were conducted with varying inlet temperatures of air (15 °C–45 °C) and air flow rates in response to the CPU temperature for air cooling. Additionally, six experiments were performed with varying inlet temperatures of PG solution (25 °C–50 °C) and airflow rates for hybrid cooling. For the 13 cases with various boundary conditions for the two methods, the CPU temperature obtained from the simulation was validated within a maximum deviation of 1.9 K as depicted in Fig. 6. This deviation occurred owing to the difficulty in maintaining consistent experimental conditions, including the inlet temperature, flow rate, and heat generation. Additionally, the assumption that the CPU heat generation is uniform across the volume introduces a potential source of error given the nonuniform and complex structure of the CPU package. Nevertheless, the maximum deviation of 1.9 K was reasonably small, and these validated simulation models were utilized in the present study. From these server models, a simulation model of immersion cooling was developed with a marginal modification in the configuration of the heatsink, inlet, and outlet as well as the replacement of the working fluid with FC-40.

The developed simulation model was utilized to investigate the thermal performance of the three cooling methods numerically at various independent variables including the CPU TDP, memory TDP, power consumption, inlet temperature, and server height. For a fair comparison among cooling methods utilizing gas, liquid, or a combination of both as working fluids, a uniform power consumption was established as the criterion for the boundary condition. This approach, based on **Eqs.** (25) and (26), was chosen instead of using identical inlet velocities or mass flow rates. The use of the inlet velocity or mass flow rate does not account for the significant differences in the thermal properties of gases and liquids, such as density and viscosity. This can result in the generation of unrealistic operating conditions for either

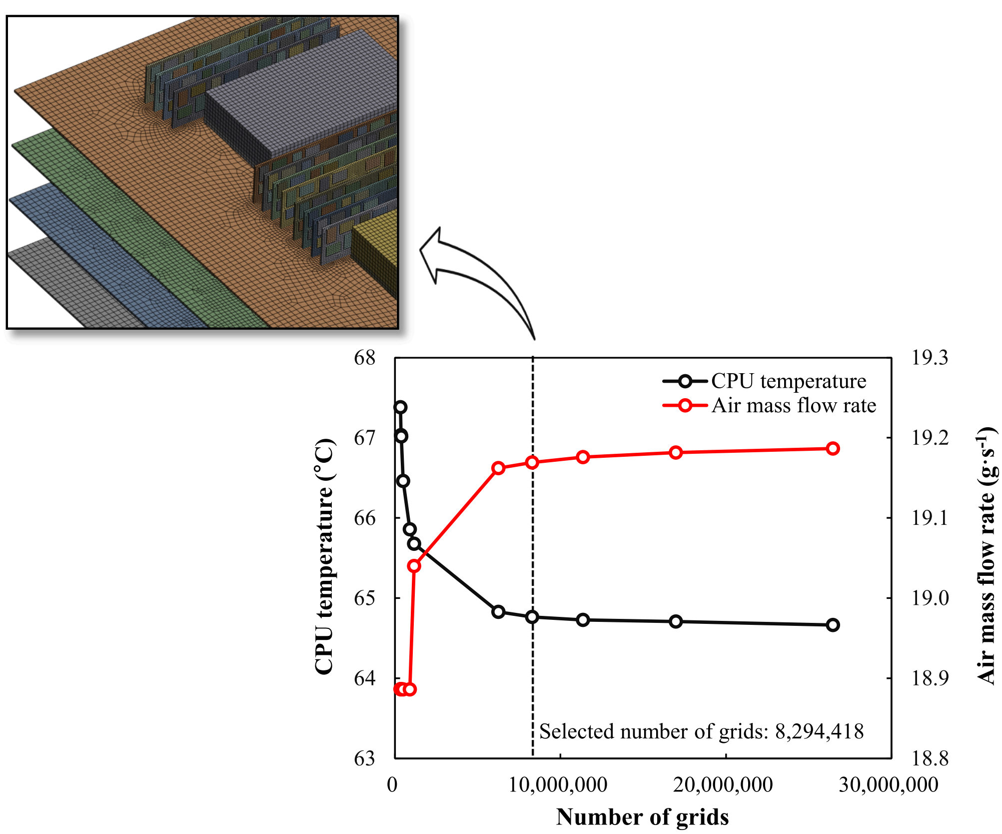

Fig. 5. Generated mesh and grid independence analysis of CPU temperature and air mass flow rate.

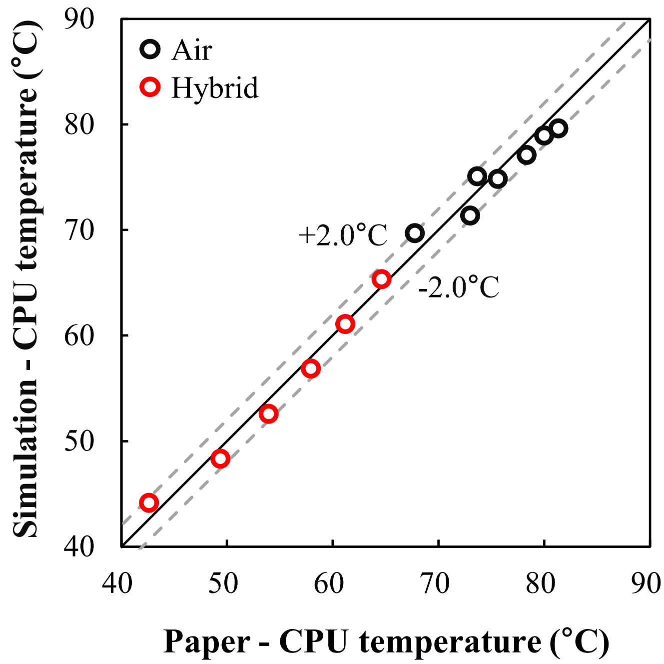

Fig. 6. Validation in CPU temperature for air and hybrid cooling.

system. In addition, a valid method for distributing fluid for hybrid cooling that uses both gas and liquid does not exist in terms of inlet velocity and flow rate. For these reasons, power consumption was chosen as the parameter that accounts for the difference in the inlet velocity and physical properties through the volumetric flow rate and pressure drop. The high density and viscosity of the dielectric fluid resulted in a large pressure drop between the inlet and outlet, which was compensated for by a lower volumetric flow rate. For hybrid cooling, the power consumption was distributed between the fans and the cold plate such that 10 % of the total power consumption was consumed by the cold plate and the remaining by the fans.

$$\mathrm{PC} = \dot{V}\Delta P_{\mathrm{fan}} \tag{25}$$

$$\mathrm{PC} = \dot{V}\Delta P_{\mathrm{pump}} \tag{26}$$

$$T_{\mathrm{cpu,avg}} = \frac{\sum T_{\mathrm{cpu}}}{n} \tag{27}$$

$$\mathrm{TR}_{\mathrm{cpu}} = \frac{T_{\mathrm{cpu,avg}} - T_{\mathrm{amb}}}{Q} \tag{28}$$

$$\mathrm{TR}_{\mathrm{memory}} = \frac{T_{\mathrm{memory,max}} - T_{\mathrm{amb}}}{Q} \tag{29}$$

$$dT_{\mathrm{cpu}} = T_{\mathrm{cpu,max}} - T_{\mathrm{cpu,min}} \tag{30}$$

$$T_{\mathrm{std}} = \sqrt{\frac{\sum \left(T_{\mathrm{cpu}} - T_{\mathrm{cpu,avg}}\right)^2}{n}} \tag{31}$$

In the context of constant power consumption, the thermal management performances of the cooling methods were evaluated using various performance indexes. The CPU temperature was measured using the average temperature of eight CPUs at their core across the server rack as calculated using **Eq.** (27). The thermal resistances of the CPU and memory were calculated using **Eqs.** (28) and (29) to evaluate the cooling performance under a range of CPU and memory TDP. The temperature difference for each CPU was calculated using **Eq.** (30), as nonuniform temperature distribution for a CPU can cause damage and unreliability [45]. The standard deviation of the CPU temperature across the server rack was calculated using **Eq.** (31) to quantify the extent to which the CPU temperature deviated from the average. This represented the homogeneity of the cooling performance across the server and the stability of server operation over an extended period.

3. Results and discussions

3.1. Effects of CPU and memory TDP

The performances of various cooling methods were investigated for a wide range of CPU TDP. Currently, a significant proportion of commercial CPUs operate at a TDP between 100 W and 200 W. Owing to the emergence of unprecedented technologies such as AI, the need for high-performance CPUs continues to grow and is likely to increase further in the future. Therefore, it is important to consider both ends of the spectrum and examine the dominant cooling technology for data centers in the future. The cooling performance of the three methods was evaluated using various parameters under identical conditions (80 W power consumption, 18 °C inlet temperature, and 4 W memory TDP).

Various performance parameters (the CPU temperature, thermal resistance of the CPU, temperature difference, and standard deviation of the temperature) for a wide range of CPU TDPs are shown in Fig. 7. As illustrated in Fig. 7**(a)**, immersion cooling is the only method that maintains the CPU temperature below the target temperature of 79 °C [46]. It exhibits the lowest value among the three cooling systems across the CPU TDP range. At a CPU TDP of 800 W, the CPU temperature for immersion cooling is 102.6 °C and 22.9 °C lower than those for air and hybrid cooling, respectively. The high heat capacity resulting from the higher mass flow rate and density, as well as the high thermal conductivity, enables immersion cooling, which cools the CPU with a dielectric fluid, to exhibit superior cooling performance compared with air cooling. Unlike immersion cooling, hybrid cooling exhibits a higher CPU temperature owing to the additional thermal resistance introduced by the indirect cooling mechanism, as suggested by Kheirabadi and Groulx (2016) [47]. In a hybrid cooling system, the CPU is cooled primarily through a cold plate rather than air. As the working fluid flows through the cold plate, it absorbs heat from the CPU through an additional layer of a copper block, which serves as a source of thermal resistance. Consequently, hybrid cooling results in higher CPU temperatures due to its inherent limitations. This phenomenon is observed more clearly by comparing the thermal resistance values presented in Fig. 7**(b)**. The average thermal resistances for immersion and hybrid cooling were relatively constant at 0.055 $\mathrm{K{\cdot}W^{-1}}$ and 0.084 $\mathrm{K{\cdot}W^{-1}}$, respectively, throughout the range. This indicates that as the CPU TDP increases, the temperature difference increases proportionally with the heat load. With high thermal conductivity and specific heat capacity, the fluid used in both immersion and hybrid cooling could effectively absorb heat from the CPU without losing its efficiency. For the constant thermal resistance, it also indicates that the convective heat transfer coefficient remains relatively stable with the fluid's thermal properties and cooling capacity not significantly affected by the increased heat load. In contrast to the other methods, the thermal resistance of air cooling varied by 15.4 % when the CPU TDP increased from 100 W to 800 W. This was significantly higher than the variations of 2.2 % and 0.6 % observed for hybrid and immersion cooling, respectively. As the temperatures of the air and CPU increase significantly due to ineffective cooling, the thermal properties of the air vary considerably, and the thermal resistance of air cooling varies accordingly.

Furthermore, the temperature deviation of air cooling was considerably higher than that of the other methods as illustrated in Fig. 7**(c)**. As the air flowed through the heatsink, the temperature rise of the air was much larger than the liquids in other cooling methods due to its low heat capacity. With higher air temperature at the later part of the heatsink channel, a large temperature difference between the front and back of the CPU appeared along with a low cooling performance of the air owing to the low convection heat transfer. This phenomenon can potentially

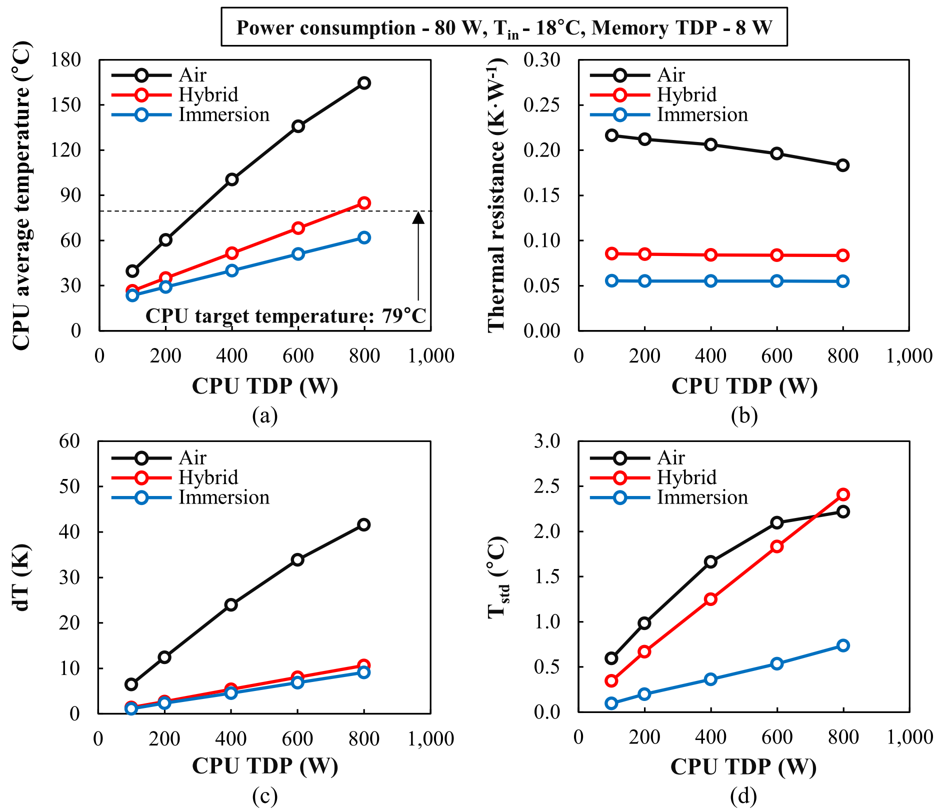

Fig. 7. Effect of CPU TDP on (a) CPU average temperature, (b) thermal resistance, (c) dT, and (d) $T_{std.}$

cause CPU damage.

Regarding the standard deviation of the CPU temperature, which represents the reliability of the cooling system across the server rack, immersion cooling demonstrated the highest stability at 800 W TDP with a standard deviation of 0.74 °C. This was 66.8 % and 69.4 % lower than those of air and hybrid cooling, respectively, as shown in Fig. 7**(d)**. The substantial disparity observed in the air cooling system can be attributed to considerable temperature variations across different servers. The temperature difference between the top and bottom CPU of the server attained 6.1 °C for air cooling, unlike the 2.1 °C observed in immersion cooling leading to a high standard deviation. This indicates that servers exert a significant influence on one another, and the location in the rack is a crucial factor. The server situated at the lowest point near the wall with no airflow below the PCB exhibited the highest CPU temperature. In contrast to air cooling, the primary source of the standard deviation for hybrid cooling was between the two CPUs from the same server. As the fluid flowed across the two CPUs in a cold plate, the working fluid absorbed heat from the first CPU. This caused a reduction in the cooling performance of the fluid for the second CPU owing to the higher temperature of the fluid. For an elevated CPU TDP, the temperature of the fluid increased significantly, resulting in a high degree of variation in the CPU temperature. Unlike air and hybrid cooling, the CPU temperature for immersion cooling exhibited a relatively consistent performance between two CPUs in the same server and across the server. The maximum temperature difference across the server was 2.1 °C, demonstrating the consistent and reliable cooling performance of immersion cooling, which is an important criterion, particularly for data centers that operate for extended periods.

The advantages of immersion cooling are depicted in Fig. 8, which illustrates the temperature contours of the three cooling methods for a CPU TDP of 800 W. The temperature of the CPU attached to the PCB is the lowest and most consistent, exhibiting a minimal gradation. In contrast, a temperature gradient is evident in the CPU for air cooling. The low heat capacity of air leads to a temperature increase as it passes through the heatsink, resulting in a clear rise in temperature for the CPUs and fins. Furthermore, the temperature rise around the heatsinks for air cooling is also depicted, in contrast to the relative uniform temperature of the working fluid for hybrid and immersion cooling throughout the server. As the temperature of the working fluid increases, it influences the thermal management of nearby components by affecting a key factor for convection, namely, the temperature difference between the heated body and fluid. Therefore, studying the thermal management of a server as a rack, rather than as a single server, is important, particularly for air cooling.

In conjunction with the CPU, memory performance is also of significant importance, given the increasing load of information processing and the speed required for processing the data. As the memory performance continues to improve, the memory TDP is also likely to increase. Consequently, the thermal management of these components is a crucial aspect. With the commercial memory TDP in the range of 4–8 W, the maximum memory TDP was selected as 20 W, considering the future development and extreme conditions. For a wide range of TDP, the maximum temperature and thermal resistance of the memory were measured for three cooling techniques, as illustrated in Fig. 9. Rather than the average temperature of the memory, its maximum temperature was selected as a more accurate representation of memory thermal management, given the significant temperature variation across the chips, as illustrated in Fig. 10. Unlike the thermal management of CPU, where the working fluid is focused on a limited number of CPUs through a heatsink and cold plate, the thermal management of memory is conducted only through the free flow of the fluid without the help of a heat dissipation enhancer. As depicted in Fig. 9, the maximum temperature of the memory at a high TDP of 20 W is lowest for immersion cooling at 44.2 °C (86.2 °C for air cooling and 113.7 °C for hybrid cooling) which is below the memory target temperature of 82 °C [48]. Unlike air and hybrid cooling, in which memories are cooled by air, immersion cooling displays the most effective cooling performance, because dielectric fluids are employed for cooling the memory. The dielectric fluid with a

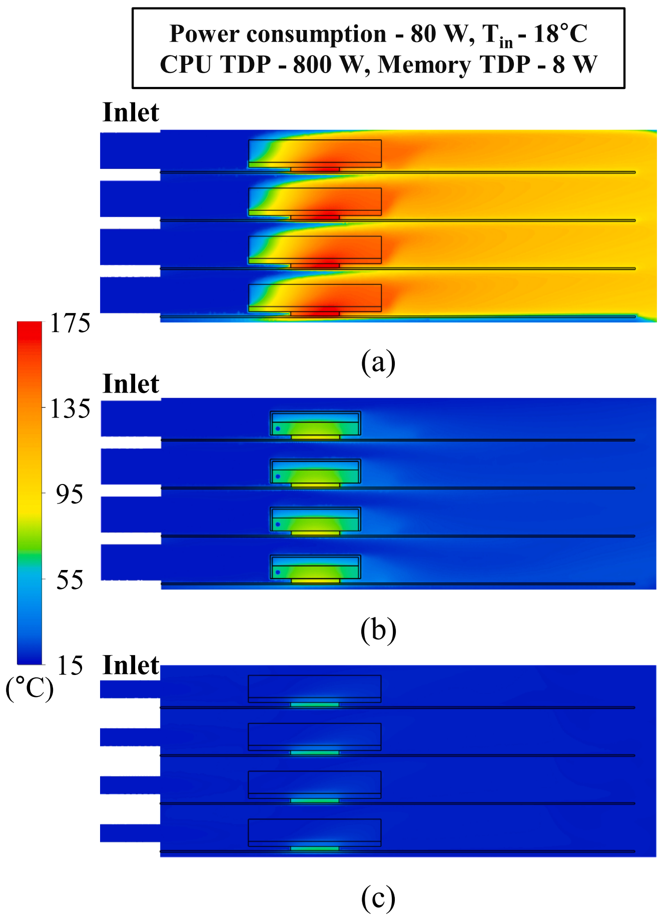

Fig. 8. Temperature contour of (a) air, (b) hybrid, and (c) immersion cooling at CPU TDP of 800 W.

high heat capacity exhibited superior cooling performance compared with air, as evidenced by the low memory temperature and thermal resistance observed in Fig. 9. In accordance with the constant thermal resistance for CPU in immersion cooling, thermal resistance for memory was relatively constant for immersion cooling as well. With the dielectric fluid's outstanding thermal properties compared with air, the cooling performance of immersion cooling was not diminished even at high memory TDP. Even considering the low temperature gradient of the memory as depicted in Fig. 10, immersion cooling has great cooling capability for the memory as well.

A comparison of the cooling performance of air and hybrid cooling revealed that the memory temperature and thermal resistance for air cooling were lower than those for hybrid cooling, except at a low memory TDP of 4 W. Because 10 % of the total power consumption was used for the cold plate during hybrid cooling, the air velocity at the inlet for hybrid cooling was lower than that for air cooling leading to higher memory temperature. However, at a low TDP of 4 W, the memory temperature for air cooling was higher than for hybrid cooling. At low memory TDP, the air cooling system exhibited a critical thermal management limitation. The warm air surrounding the CPU and heatsink, characterized by high localized temperatures, directly compromised memory temperature regulation. This resulted in a higher memory temperature for air cooling than for hybrid cooling, even with a higher air flow rate. The exceptionally high thermal resistance for air cooling at 4 W TDP is unlike the relatively constant thermal resistance for the other cooling methods across the range of memory TDP. This indicates that external factors such as a high heatsink temperature affect the memory temperature as demonstrated by the extraordinary value, which is not in line with the trend.

3.2. Effect of server power consumption

The temperatures of the CPU and memory, in conjunction with the temperature difference of the CPU, at various power consumption levels between 20 W and 100 W are displayed in Fig. 11. In addition to maintaining a temperature below the target temperature, the other main goal of thermal management in a data center is to minimize the power consumption and reduce the operational costs. Through this investigation, a guideline for the appropriate power consumption of each cooling method considering both component temperature and energy consumption was provided. For the low and high power consumption values, 20 W and 100 W were selected. For a range of power consumption levels, the inlet fluid temperature was 18 °C, and the CPU and memory TDPs were set at 200 W and 8 W, respectively.

At a low power consumption of 20 W, the CPU temperature of air, hybrid, and immersion cooling were 140.1 °C, 36.9 °C, and 29.8 °C, respectively; at a high power consumption of 100 W, these were 49.4 °C, 34.7 °C, and 28.9 °C, respectively. At a power consumption of 20 W, immersion cooling demonstrated the smallest CPU temperature difference of 2.4 K, which was 10.1 K and 0.5 K lower than that of air and hybrid cooling, respectively. Throughout the range of power

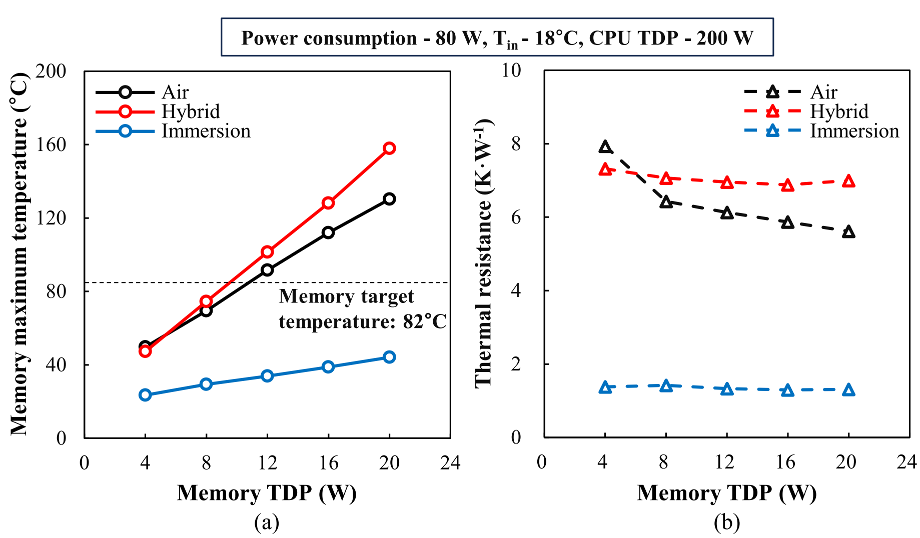

Fig. 9. Effect of memory TDP on (a) maximum temperature and (b) thermal resistance.

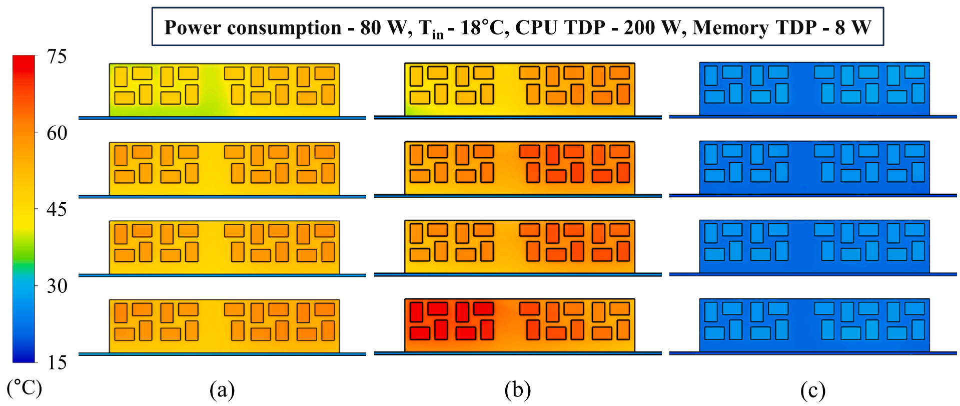

Fig. 10. Temperature contour of the memory for (a) air, (b) hybrid, and (c) immersion cooling at memory TDP of 8 W.

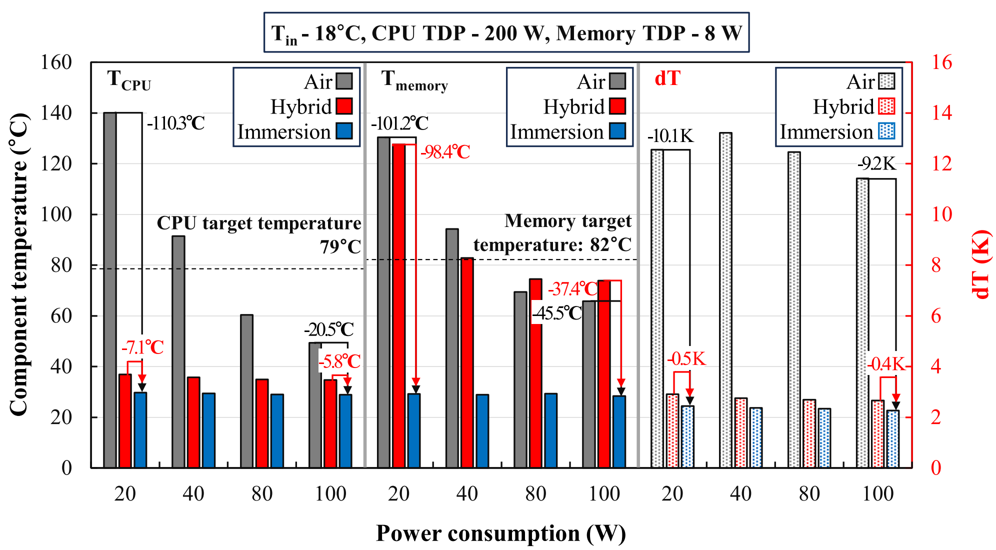

Fig. 11. Effect of power consumption on component temperature and temperature difference.

consumption, immersion cooling exhibited the most effective cooling performance for the CPU, with minimal fluctuations in temperature. Hybrid cooling, also cooled by liquid, showed limited fluctuations. Unlike immersion and hybrid cooling, air cooling, where the CPU is cooled by air, showed a notable increase in CPU temperature as power consumption decreased. The power consumption is calculated using the volume flow rate and pressure drop across the fan or pump. Therefore, the physical properties of the working fluid play a significant role in determining the power consumption. For liquids such as the dielectric fluid used for immersion cooling and the PG solution used for hybrid cooling, the density and viscosity are considerably higher than that of air. This results in higher fluctuations in the pressure drop as the inlet velocity varies. Therefore, the inlet velocity at a low power consumption is not significantly lower than that at a high power consumption for hybrid and immersion. In contrast, a significantly lower inlet velocity for air cooling at low power consumptions of 20 and 40 W led to failure in satisfying the target CPU temperature. A similar trend was observed for memory cooling. Unlike immersion cooling, which employs a dielectric fluid, air and hybrid cooling utilize air, resulting in a large variation in memory temperature at a low power consumption. At a low power consumption of 20 W, the memory temperatures for air, hybrid, and immersion cooling were 130.4 °C, 127.6 °C, and 29.2 °C, respectively. Immersion cooling is the only cooling method that satisfies the memory target temperature of 82 °C. In the context of memory cooling at a low power consumption, it is necessary to consider the role of natural convection. It originates from the motion of a fluid driven by its density difference at different temperatures. This phenomenon is distinct from forced convection, which is caused by the motion of a fluid induced by an external force from fans and pumps. As the fluid velocity decreases at a low power consumption, a region of stagnant flow or “dead region” forms around the memory. In this region, natural convection becomes a more prominent method of cooling, and a dielectric fluid with a high heat capacity and thermal conductivity can effectively cool the memory. This results in a remarkable performance even at a low power consumption.

3.3. Effect of coolant inlet temperature and server height

The thermal management of server components at various fluid inlet temperatures was investigated, considering the liquid and air cooling classification as defined by the American Society of Heating, Refrigerating and Air-Conditioning Engineers [49]. Although a low inlet fluid temperature in data centers can result in a low component temperature, there are numerous challenges associated with the management of low temperature fluids. In terms of supplying fluid at low temperatures, additional energy is required for air conditioning or refrigeration. This results in an inefficient use of resources when considering the management of the data center building. In terms of the returning fluid, retrieving heat from a low temperature fluid is significantly more challenging than removing heat away from a high temperature fluid, which also results in inefficiency. Therefore, it is preferable to operate a data center using working fluids at high temperatures for efficiency and convenience of management.

In this study, six temperatures were selected from the liquid- and air-cooling classifications to assess the performance of each cooling method at high fluid temperatures. The temperature of each working fluid for the cooling methods was maintained constant, and the performance was assessed at identical power consumption, CPU TDP, and memory TDP. As the temperature of the inlet fluid increased, the temperatures of CPU and memory increased linearly. This indicated a minimal variation in thermal management performance in terms of thermal resistance as shown in Fig. 12 as the variation in fluid properties was limited. The target temperature for the components was considered to examine whether the target could be achieved even at a high inlet temperature of 45 °C. The target temperature of 79 °C for the CPU could not be achieved with air cooling, as the CPU temperature was 86.85 °C at an inlet air temperature of 45 °C. Similarly, a target temperature of 82 °C for memory could not be obtained with air and hybrid cooling with a memory temperature of 94.1 °C and 101.0 °C, respectively. As stated previously, cooling methods utilizing air (air cooling for CPU and memory and hybrid cooling for memory) with suboptimal heat dissipation capabilities as the working fluid demonstrated limitations at high inlet temperatures. These outcomes substantiated the superiority of immersion cooling as the most effective cooling method even at a high inlet temperature of 45 °C. These further emphasized the necessity for air and hybrid cooling to be operated within specific boundaries for the efficiency of the system and the temperature of the components.

In addition to inlet temperature, the effect of server height on the cooling methods was investigated. Considering the configuration of the server model used for validation, a server height of 1 U was selected as the baseline for previous studies. The thermal management performance of different cooling methods for CPU and memory was studied for the range of server heights from 0.8 U to 1.2 U. The CPU temperatures at different server heights are depicted in Fig. 13(a), indicating a limited increase in temperature as the server height increased. The temperature of the CPU for air, hybrid, and immersion cooling increased by 4.0°C, 0.62 °C, and 0.29 °C, respectively, as the server height varied from 0.8 U to 1.2 U. Unlike the CPU, where the variation was minimal, the memory temperature for air, hybrid, and immersion cooling increased by 24.8 °C, 25.4 °C, and 2.7 °C each. It is important to note the large temperature variations in the memory for air and hybrid cooling, as shown in Fig. 13 **(b)**. As the server height increased, the temperatures of both CPU and memory increased because of the bypass effect of the fluid, in accordance with Luo et al. (2024) [47]. Our results expand upon the work of Luo et al. (2024) as indicated by the varying magnitude of difference for different cooling methods and components. Given a relatively constant volume flow rate at the inlet for different server heights, it is optimal for the flow to be concentrated in the CPU and memory in terms of the thermal management of these components. However, as the server height increased, there was more space for the fluid to bypass, and the average flow rate of the working fluid around the components decreased. This increased in the component temperature. The extent of the bypass effect on the CPU was limited, because the fluid inlet was focused on the heatsink attached to the CPU. Furthermore, the velocity of the fluid entering the heatsink was not affected significantly by the server height. However, the maximum memory temperature for air and hybrid cooling, where the memory was cooled by air, was affected significantly by the server height. This demonstrated the considerable impact of the bypass effect on memory cooling. As the memory of the server is distributed across the server, it is not feasible to locate a fluid inlet for each memory. Consequently, a region with a low fluid velocity was created. In this region, the enhanced bypass effect for a larger server height caused a larger variation in the memory temperature than in the CPU temperature for air cooling. The bypass effect was limited for immersion cooling owing to the high cooling performance even at low flow velocity. It is crucial to note that only enlarging the server has a detrimental impact on the cooling performance of the CPU and memory due to the bypass effect. To simultaneously increase the size of the server and enhance the thermal management performance under extreme conditions, it is necessary to install additional instruments such as larger heatsinks and guide vanes for memory to prevent the bypass effect.

3.4. Cooling performance of various dielectric fluids

A comparison of cooling methods under diverse operating conditions including a high TDP of electronic components revealed that immersion cooling, which employs FC-40 as a working fluid, exhibited the most effective cooling performance in terms of the component temperature, temperature difference, and standard deviation of the CPU temperature across the server. For further analysis of immersion cooling, the impact of the dielectric fluids listed in Table 5 on the temperature of the CPU and memory was investigated. To facilitate the comparison under more

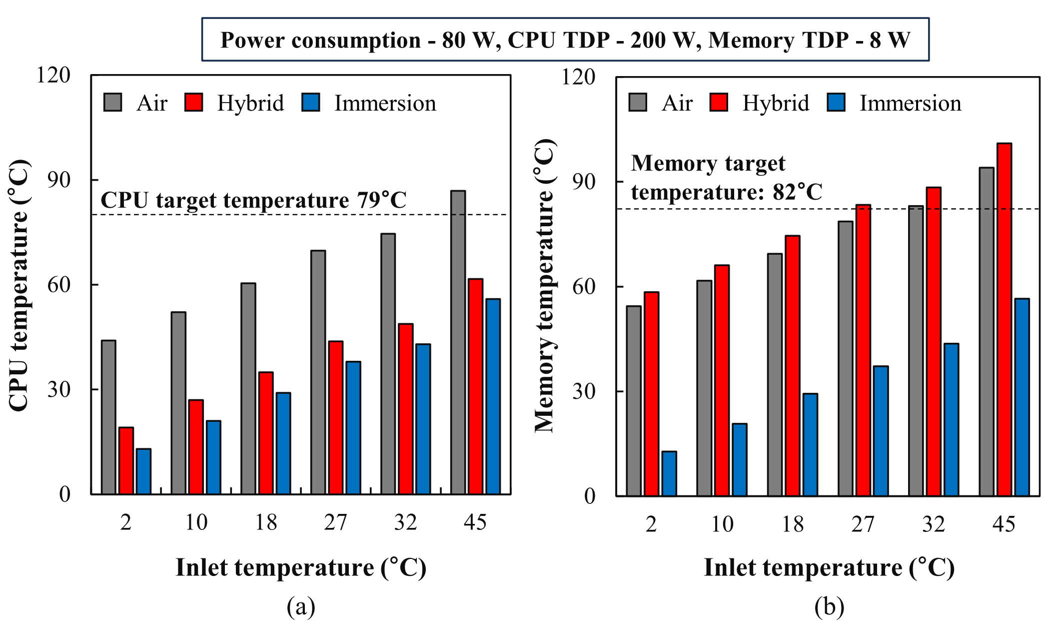

Fig. 12. Effect of inlet temperature on (a) CPU and (b) memory temperature.

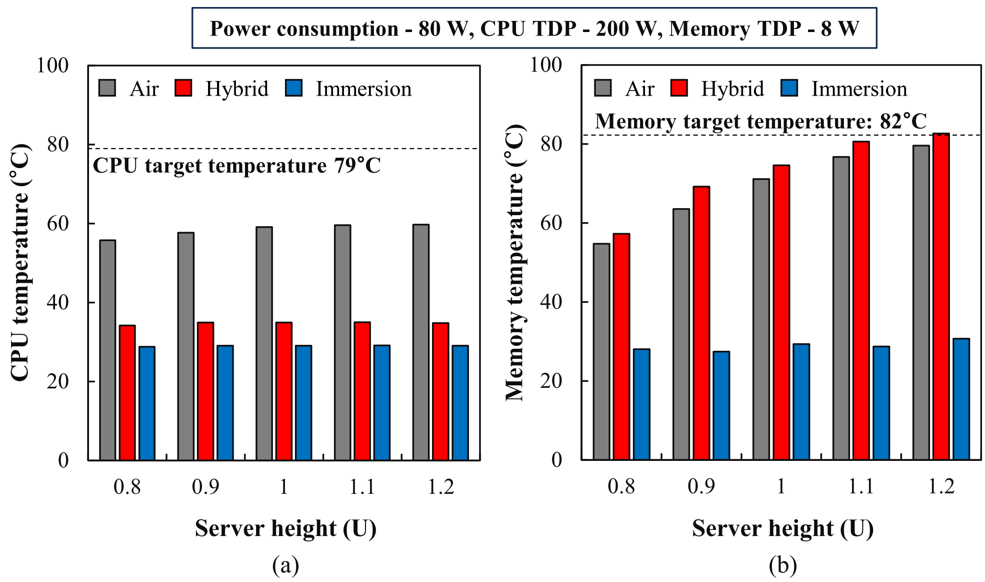

Fig. 13. Effect of server height on (a) CPU and (b) memory temperature.

rigorous conditions, the TDP of the CPU and memory were set at 800 W and 20 W, respectively. These represented the highest TDP values from earlier studies. The temperatures of the CPU and memory were measured at different power consumption levels of 20 and 100 W for comparison in distinct states.

Fig. 14 **(a)** depicts the temperature of the CPU at a high TDP of 800 W for nine dielectric fluids. Among the nine fluids, TMC-7200 with the lowest viscosity exhibited the lowest CPU temperature of 61.7 °C at a power consumption of 20 W. Opteon SF 10 and Novec 7500 also with low viscosity demonstrated comparably low CPU temperatures. The

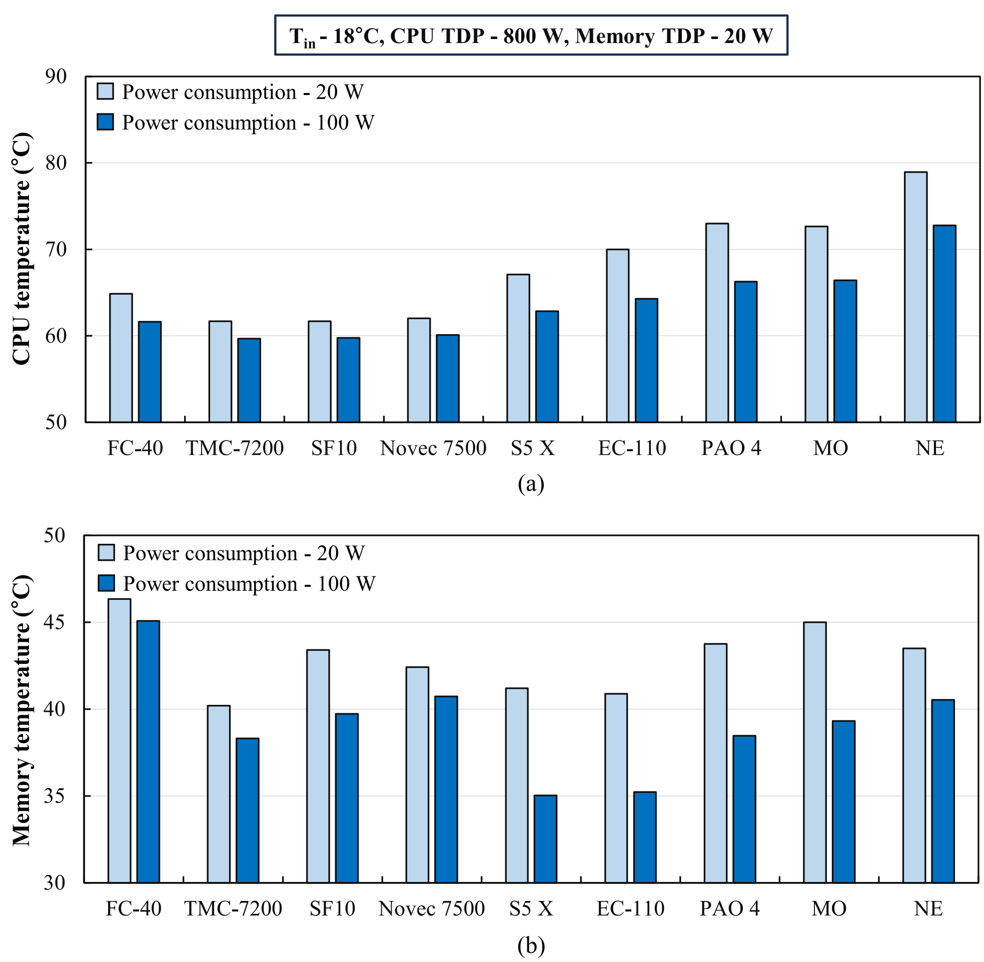

Fig. 14. (a) CPU and (b) memory temperature for various dielectric fluids.

fluid with the highest viscosity, natural ester (NE), exhibited the highest CPU temperature of 78.9 °C which was 17.2 °C higher than that of TMC-7200. These findings highlight the critical role of viscosity in CPU cooling. Our results expand upon the work of Chen et al. (2024), who stated the importance of viscosity without the consideration of heatsink [50]. The significance of viscosity is also portrayed through the analysis of pressure drop across the heatsink for various dielectric fluids, as shown in Fig. 15. The pressure drop, calculated using **Eq. (6)−(16)**, accounts for the combined effects of fluid viscosity, fluid density, and the heatsink configuration. For CPU cooling, the dielectric fluid, especially one with high viscosity, must flow through narrow channels between the heatsink fins. As the fluid flows, it encounters significant frictional force along the fin surfaces, which creates resistance to the flow. This resistance increases with higher fluid viscosity, as the frictional force rises. Consequently, the pressure required to maintain adequate flow increases, directly impacting the cooling performance of the CPU. Such a phenomenon is depicted in the static pressure contour of natural ester and TMC-7200 in Fig. 16. For the pressure contour of natural ester with high viscosity in Fig. 16**(a)**, the high-pressure zone appears in front of the heatsink where the static pressure builds up due to the resistance of the flow. As the natural ester struggles to flow through the narrow channel of the heatsink, the velocity of the fluid decreases suddenly and creates a semi-stagnant region. In contrast to the natural ester, TMC-7200 does not create such a region with a relatively consistent pressure contour indicating a well-established flow of the fluid, as shown in Fig. 16**(b)**. This suggests that TMC-7200 experiences less resistance through the channel of the heatsink and shows higher cooling performance for the CPU.

In contrast, the performance trends of the dielectric fluids differed for memory cooling, as illustrated in Fig. 14**(b)**. At a power consumption of 20 W, TMC-7200 exhibited the lowest memory temperature of 40.2 °C, whereas FC-40 demonstrated the highest memory temperature of 46.3 °C. Unlike the CPU cooled with a heatsink, the memory was cooled through a free flow of fluid over the surface of the memory in the open area. For the memory chip with the highest temperature located far from the inlet, the fluid flowed significantly slowly, and a mixture of natural and forced convection played a role. Given the variability in the location of the maximum memory temperature and the multitude of parameters that can influence the cooling performance, it is challenging to identify a single variable that can be used to predict the memory cooling performance. Nevertheless, fluids with high thermal conductivity displayed relatively lower memory temperatures, suggesting a key role in the thermal management of the memory. Considering the cooling performance of various dielectric fluids for CPU and memory, it would be crucial to evaluate the cooling performance of both the CPU and memory independently when selecting an optimal dielectric fluid for immersion cooling.

4. Conclusions

This study conducted a comprehensive analysis of immersion cooling with air and hybrid cooling methods for a high-energy density server rack with a high heat generating electronic components. With the specific characteristics of the server rack in consideration, the thermal management performance of cooling methods was evaluated under various operating conditions, including CPU and memory TDP, power consumption, inlet temperature, and server height. In addition, immersion cooling methods using various dielectric fluids were investigated with a focus on the thermal and hydraulic performance of the heatsink. The significant results of this current study are summarized below:

- At a high CPU TDP of 800 W, immersion cooling was the only cooling method to achieve the CPU target temperature of 62.0°C. This was 102.6 °C and 22.9 °C lower than the temperatures achieved by air cooling and hybrid cooling. In terms of thermal resistance, temperature difference, and standard deviation, immersion cooling was the most effective.
- As the sole method for cooling memory through dielectric fluid rather than air, immersion cooling exhibited the lowest memory temperature at 20 W TDP with 44.2 °C while the memory temperatures for air and hybrid cooling were 86.2 °C and 113.7 °C, respectively.
- At different power consumptions and server heights, immersion cooling was the least affected by the varying conditions with the lowest component temperature. As the server height increased, the temperature of the components increased because of the bypass effect, which was more pronounced in cooling methods that employed air.
- The impacts of the various dielectric fluids on the component temperature for immersion cooling were evaluated with TMC-7200 with the lowest viscosity emerging as the most effective for CPU cooling. The performance trends for cooling the CPU and memory exhibited distinct responses to the thermal properties.

The study illustrated the notable thermal efficiency of immersion cooling, even under harsh operating conditions. However, certain limitations remain, including the computational nature of the results and

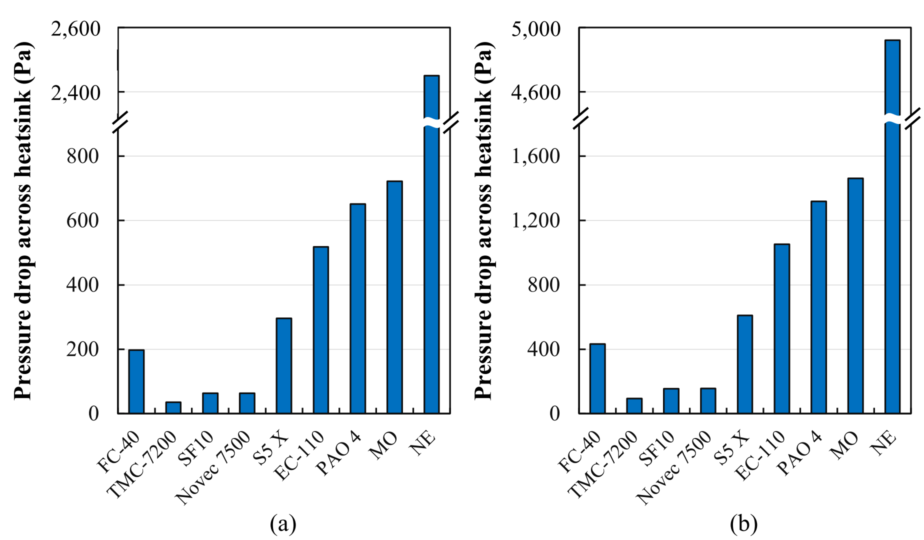

Fig. 15. Pressure drop across heatsink at the velocity of (a) 0.1 m·s^{-1} and (b) 0.2 m·s^{-1}.

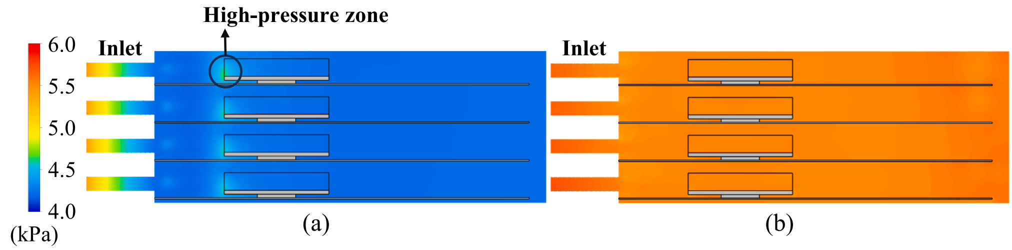

Fig. 16. Pressure contour of (a) natural ester and (b) TMC-7200 at 20 W power consumption.

the lack of consideration for state-of-the-art advancements in each cooling system, which could limit the generalizability of the findings.

Future work could overcome these limitations by integrating novel advancements into the existing model. As for immersion cooling, passive enhancements such as heat pipes, novel heatsink designs, and vapor chambers could be incorporated to improve performance. Additionally, active systems, including targeted immersion cooling and jet liquid immersion cooling, could be explored. Considering the additional power consumption required for the active system, a control logic system that dynamically regulates additional cooling based on the heat generation electronic components could be developed for more powerful and efficient thermal management, as the CPU TDP increases further.

To conclude, immersion cooling represents a promising solution for data center cooling and offers broad applicability to other high heat generating electronic systems, such as energy storage systems. As cooling requirements and power consumption continue to rise, immersion cooling holds significant potential for addressing these challenges in future electronic cooling.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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